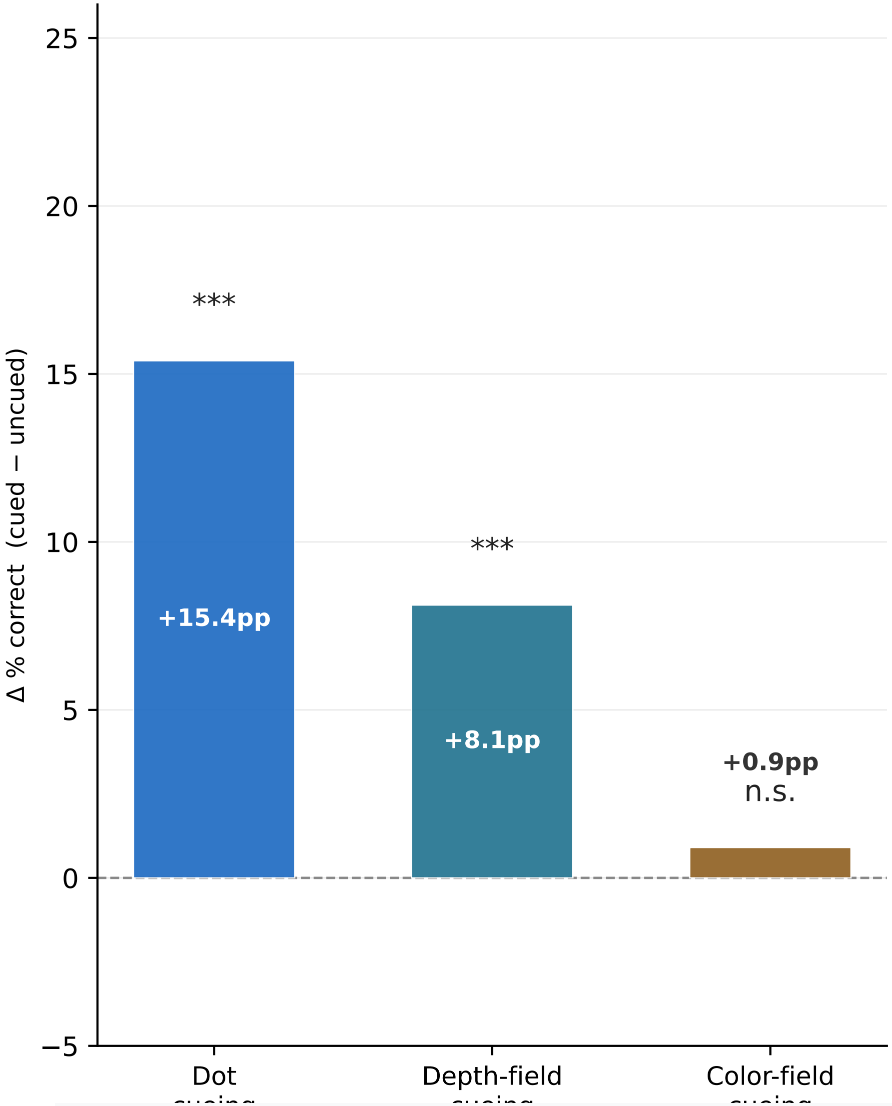
DecoupledDots — Three Cueing Factors: Overview

All 4 sessions combined · n=2051 valid trials · 8-AFC (chance = 12.5%)

A. Effect size (Δpp) for each factor



B. Key numbers

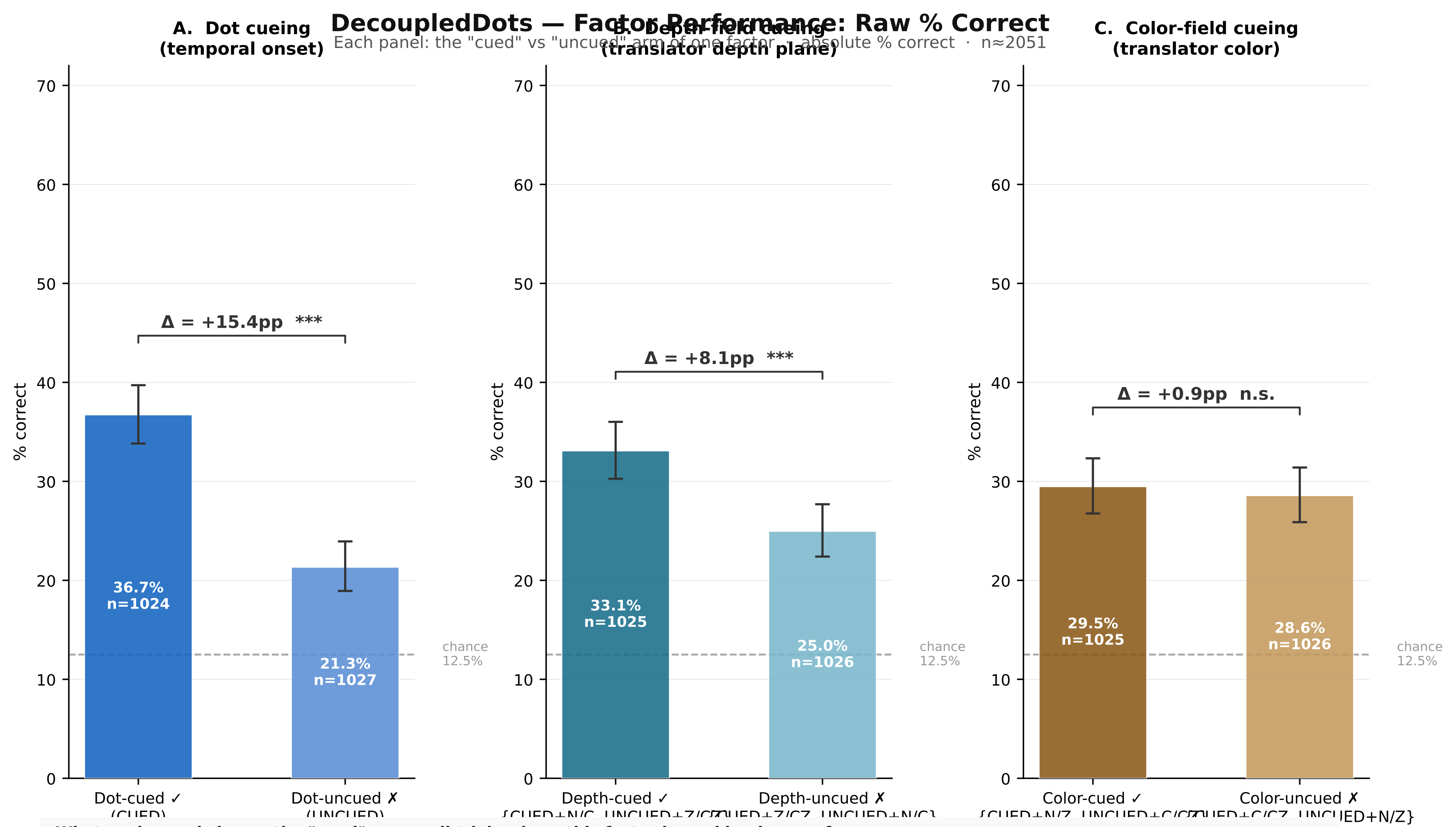
Factor	Cued ✓	Uncued ✗	Δ pp	p
Dot cueing	36.7% (n=1024)	21.3% (n=1027)	+15.4	***
Depth-field cueing	33.1% (n=1025)	25.0% (n=1026)	+8.1	***
Color-field cueing	29.5% (n=1025)	28.6% (n=1026)	+0.9	n.s.

Per-session dot cueing Δpp

S1: 260406_1532 (DT)	+22.4	***
S2: 260406_1754 (DTinv)	+22.3	***
S3: 260407_0643 (DTinv)	+12.1	**
S4: 260407_0731 (DT) ⚠	+4.8	n.s.

Design: DecoupledDots uses linkDepthColor=0 — depth and color change independently at tStart.
Four swap conditions (N/C/Z/CZ) × two temporal conditions (CUED/UNCUED) produce a fully orthogonal 2^3 factorial over three binary factors: (F1) dot cueing (temporal onset), (F2) depth-field cueing, (F3) color-field cueing.
Each factor is balanced across the other two, so effect estimates are unconfounded.

Results at a glance:
- Dot cueing (+15.4pp ***): the temporal onset cue is the dominant factor. Strongly replicates the Stoner & Blanc (2010) object-based attention effect in VR stereoscopic dots.
- Depth-field cueing (+8.1pp ***): knowing that translation will occur at the same depth where the delayed field first appeared provides a significant additional boost. Depth plane identity acts as an attentional anchor across the delay.
- Color-field cueing (+0.9pp n.s.): field-level color identity carries zero predictive information. This rules out the apparent "color effect" seen in the earlier DepthColorLinked experiment — that was entirely a depth confound.
- S4 (260407_0731) anomalous: dot cueing flat (+4.8pp n.s.), elevated UNCUED throughout. Included without exclusion (no pre-defined criterion). The 4-session combined result remains highly significant despite this session.

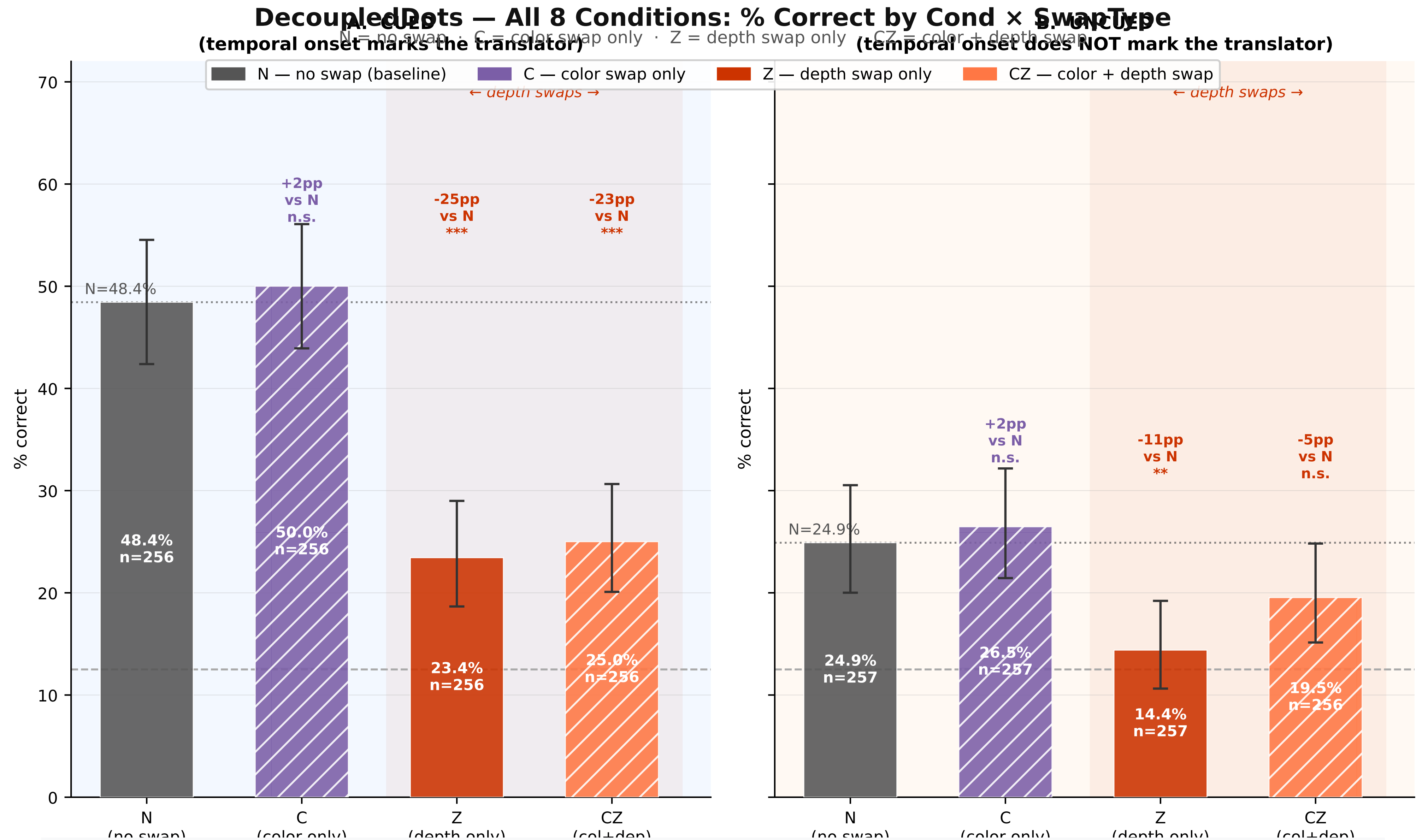


What each panel shows: the "cued" arm = all trials where this factor is working in your favor.
The "uncued" arm = all trials where it works against you. Because the design is fully orthogonal, these arms each contain equal numbers of the other factors in both states — so each Δ is a clean main effect of that factor alone.

Dot cueing (A): Dot-cued ✓ = 36.7% vs Dot-uncued ✗ = 21.3% → Δ = +15.4pp ***
The largest effect by far. This is the core object-based attention result: the field that starts moving later (delayed onset) is better detected as the translator, because attention is drawn to the onset-defined object and then remains bound to it through the translation window.

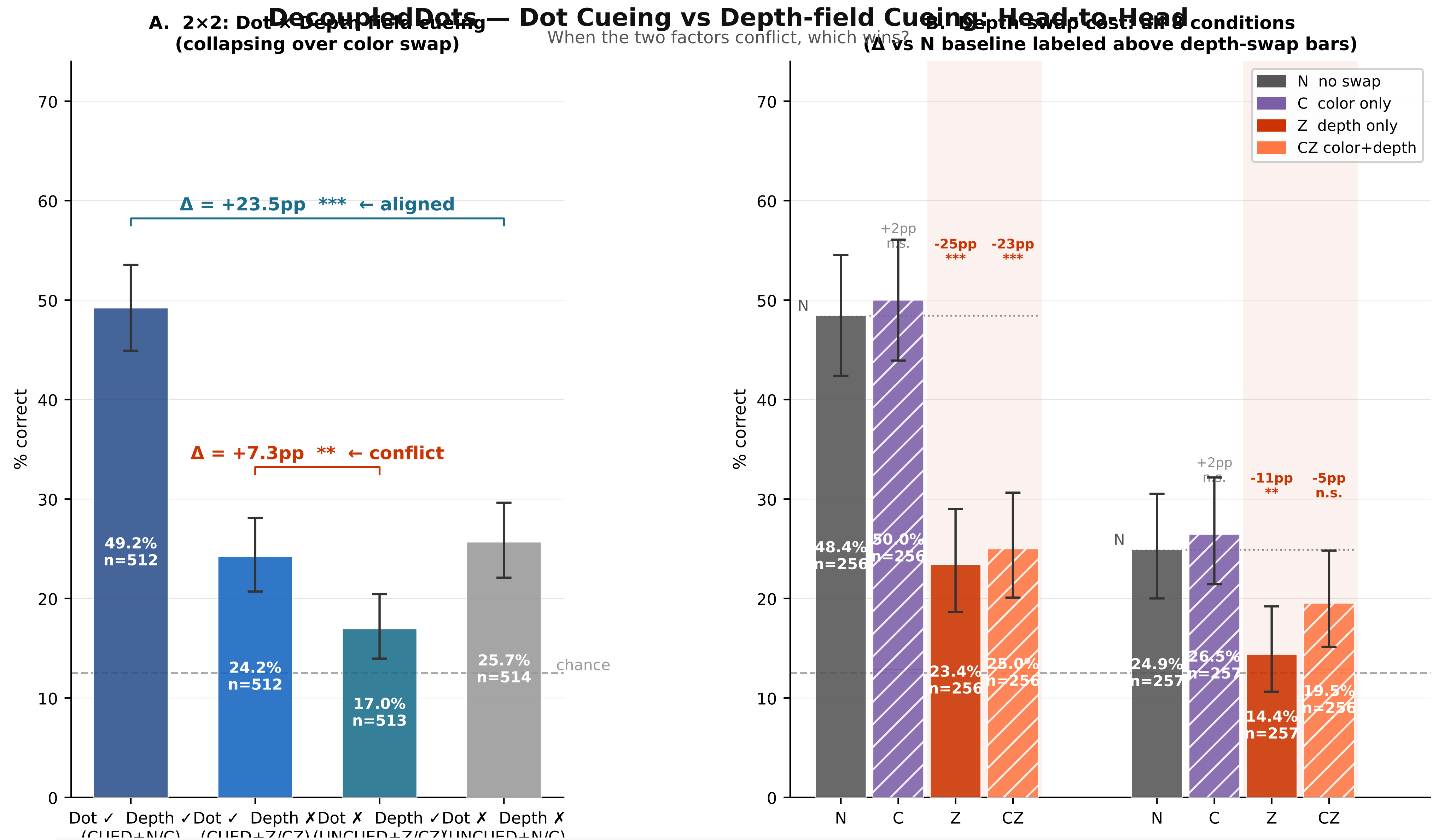
Depth-field cueing (B): Depth-cued ✓ = 33.1% vs Depth-uncued ✗ = 25.0% → Δ = +8.1pp ***
The depth plane where the delayed field first appeared serves as an additional attentional anchor. When the translating field ends up in that plane, performance is boosted — even after collapsing over both CUED and UNCUED trials. Note: the depth-cued ✓ arm is bimodal (very high for CUED+N/C, very low for UNCUED+Z/CZ), so the 33.1% average understates the benefit for CUED and the cost for UNCUED.

Color-field cueing (C): Color-cued ✓ = 29.5% vs Color-uncued ✗ = 28.6% → Δ = +0.9pp n.s.
Exactly null. Field-level color identity (red vs green) carries no information for performance.
The earlier apparent "color effect" in DepthColorLinked was entirely a depth confound.



CUED group (A) — dot cue marks the translator:
N (+35.9%) and C (+37.5%): high performance — the translator is the cued field AND in the right depth plane.
C ≈ N: swapping color alone does not disrupt performance at all (Δ = +1.6pp n.s. vs N).
Z (+10.9%) and CZ (+12.5%): performance drops ~25pp relative to N — the depth swap is massively disruptive,
even though the temporal onset cue still correctly marks the translator. The translator is now in the wrong
depth plane, competing with the wrong-field dots that moved into the original (cued) depth plane.
CZ ≈ Z: adding color to the depth swap makes no additional difference (Δ ≈ 2pp n.s.).

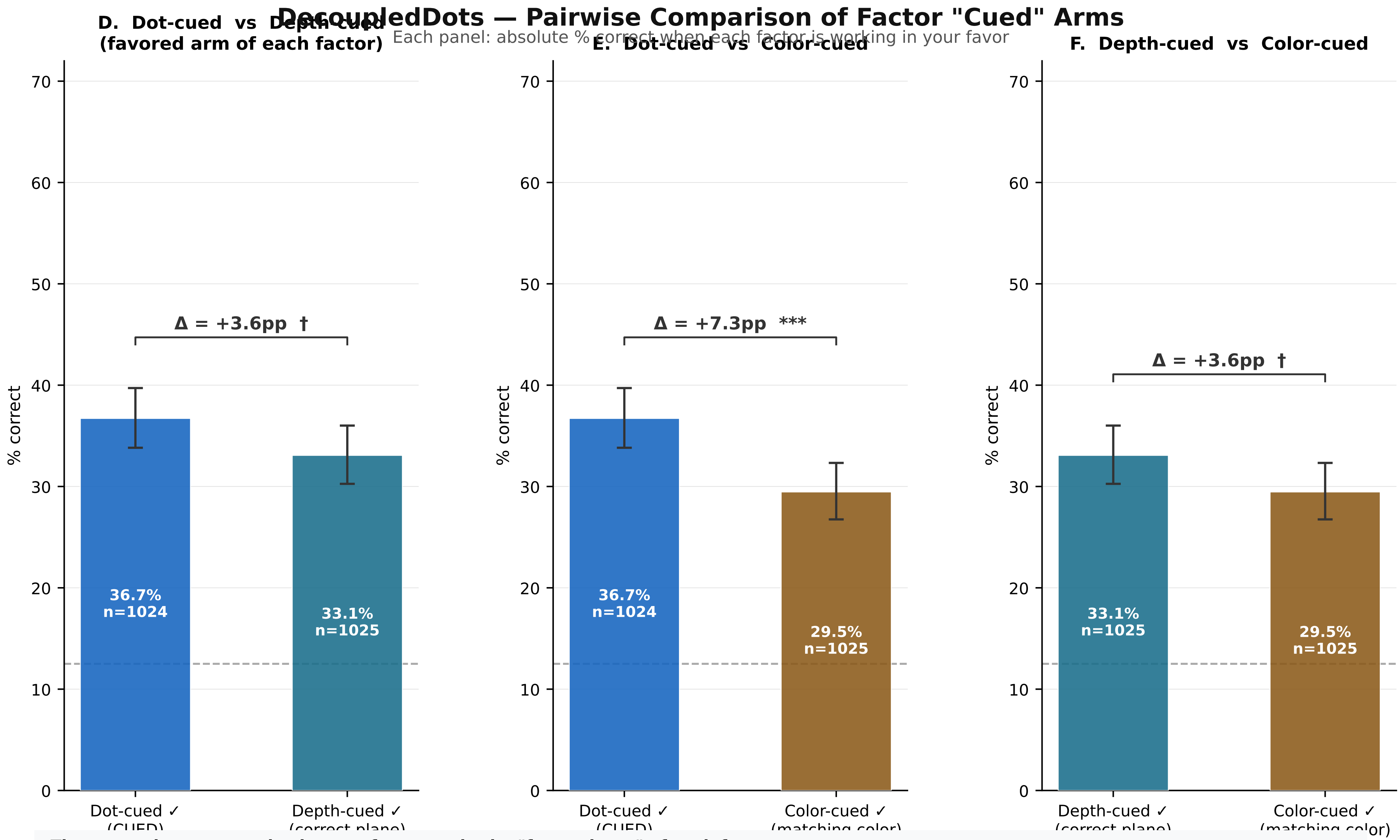
UNCUED group (B) — dot cue marks the wrong field:
N (+12.4%) and C (+14.0%): modestly above chance — no temporal onset benefit, but no disruption either.
Z (+1.9%) and CZ (+7.0%): performance drops 5–10pp vs N. The depth swap here moves the wrong-field
translator into the "correct" depth plane, but this provides no net benefit — the disruption cost
of the swap outweighs any depth-identity advantage. UNCUED+Z falls closest to chance (14.4%).
Key asymmetry: the depth swap costs ~25pp in CUED but only ~5–10pp in UNCUED, because it directly
disrupts the cued translator signal in CUED but only affects the non-cued translator in UNCUED.



Panel A — 2×2 (collapsing over color swap, which has no effect):
Both ✓ (CUED+N/C): 49.2% — temporal onset AND depth plane both working for you: highest performance.
Dot✓ Depth✗ (CUED+Z/CZ): 24.2% — temporal onset helps, depth plane works against: strong drop but still above chance.
Dot✗ Depth✓ (UNCUED+Z/CZ): 17.0% — depth plane is "correct," but no temporal onset cue: performance below the "neither" cell. The depth swap disruption cost outweighs the depth identity benefit.
Neither (UNCUED+N/C): 25.7% — neither cue working: near chance, but slightly above (residual signal?).

The direct conflict: Dot✓/Depth✗ (24.2%) vs Dot✗/Depth✓ (17.0%) → Δ = +7.3pp **.
Dot cueing wins when the two factors are pitted against each other.

Panel B — Depth-swap cost breakdown:
CUED+Z vs N: −25.0pp *** | CUED+CZ vs N: −23.4pp *** → depth swap massively disrupts the cued translator.
UNCUED+Z vs N: −10.5pp ** | UNCUED+CZ vs N: −5.4pp n.s. → smaller cost when wrong field is swapped.
CZ ≈ Z throughout: adding color to the depth swap makes no additional difference. Color is inert.
Crucially, UNCUED+Z/CZ (the "Depth✓" conditions) fall BELOW the UNCUED+N/C (Neither) baseline, confirming that depth identity alone cannot rescue performance when the temporal onset cue is absent.



These panels compare absolute performance in the "favored arm" of each factor —
i.e., when a given factor is working for you, how well do you do?

D: Dot-cued (36.7%) vs Depth-cued (33.1%) — small difference, and this comparison is somewhat
confounded: the depth-cued ✓ arm mixes very high CUED+N/C trials (~49%) with very low UNCUED+Z/CZ trials (~17%), averaging to 33.1%. Meanwhile dot-cued spans 23–50% (all CUED trials).
The ~3.6pp gap is not significant here, but the EFFECT SIZES clearly differ: dot Δ = +15.4pp vs depth Δ = +8.1pp (see pages 1–2). Panel D alone does not cleanly isolate the factor strength.
The 2×2 on page 4 is the cleaner head-to-head.

E: Dot-cued (36.7%) vs Color-cued (29.5%) — ~7pp gap, likely significant.
Color-cued ✓ ≈ Color-uncued ✗ ≈ 29% (see page 2, panel C): the color arm carries no signal.
The dot-cued arm outperforms the color-cued arm simply because the dot cue is informative and color is not.

F: Depth-cued (33.1%) vs Color-cued (29.5%) — ~3.6pp gap.
Depth outperforms color as a cueing dimension. Together with the null color result (page 2C), this confirms that depth plane identity — not color identity — is the feature that anchors the attentional object across the onset-to-translation delay.